Functional Requirements:-

Student Function

→ user Registration / login

→ Attempt Assessment.

→ View Questions

→ writing & submitting Codes.

→ View Submission status.

→ View Results.

Admin Functions:-

→ Create ~~function~~ Assessment.

→ Manage Questions.

→ Manage Test Cases.

→ Review Results.

Non Functional Requirements:-

→ Availability.

→ Scalability.

→ Security.

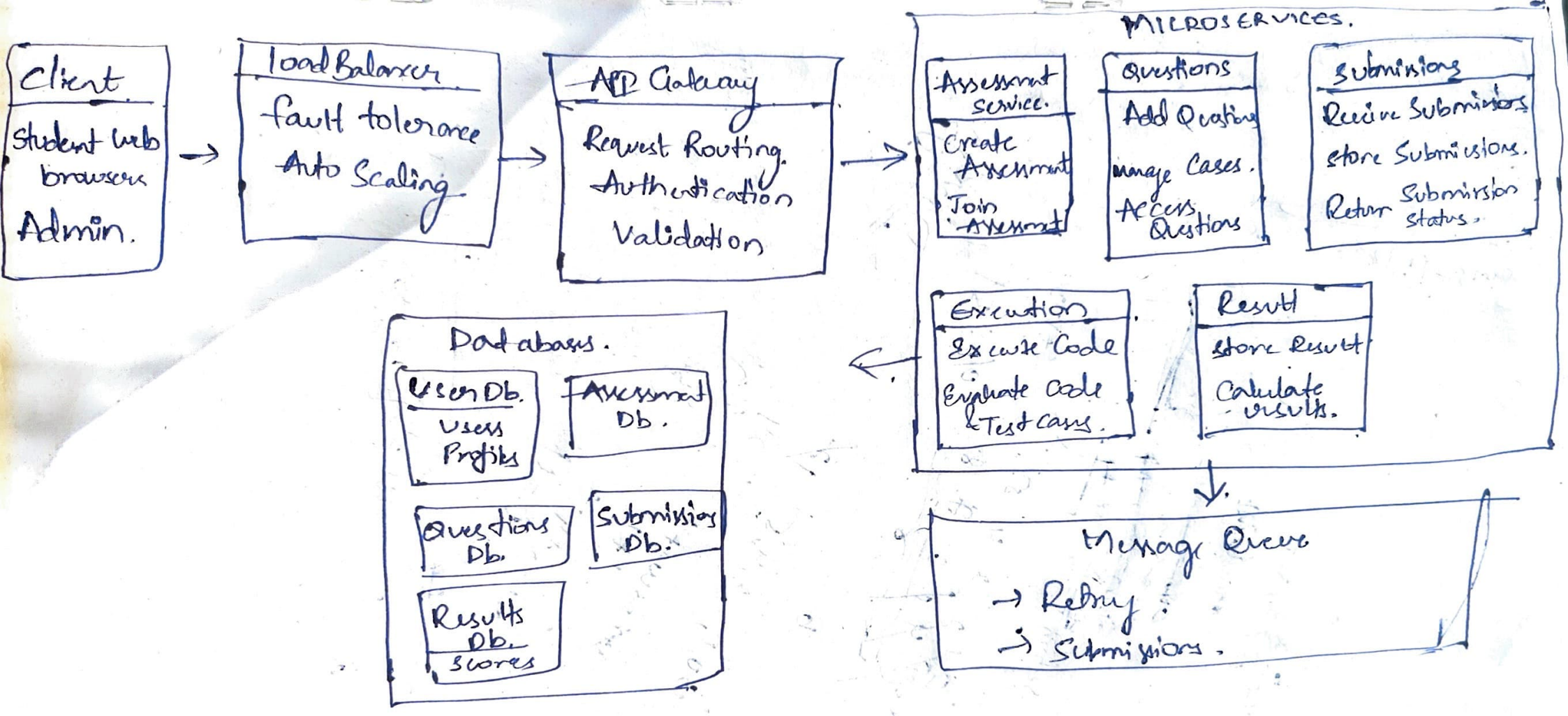
→ Reliability

→ API Response time.

→ Fault Tolerance.

Assumptions:

Users: 1) Has more no. of registered students who attempt the Exam at same time


Client
Student Web browser
Admin.
Load Balancer
fault tolerance
Auto Scaling
API Gateway
Request Routing.
Authentication
Validation
MICROSERVICES.
Assessment service.
Create Assessment
Join Assessment
Questions
Add Question
manage Cases.
Access Questions
Submissions
Receive Submissions
store Submissions.
Return Submission status.
Execution
Execute Code
Evaluate code & Test cases.
Result
store Result
Calculate results.
Databases.
User Db.
Users Profiles
Assessment Db.
Questions Db.
Submissions Db.
Results Db.
scores
Message Queue
→ Retry
→ Submissions.